HDOC Day-45 - C Programming

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Batches: 15 & 16

Aim

To write a menu-driven C program using separate functions to change a date from **dd/mm/yyyy** to **dd-mmm-yy** format and to print all possible contiguous sub-strings of a given string.

Operations

1	Change date format: **12/10/2026 -> 12-Oct-26**
2	Print all sub-strings. Example for **abc**: a, ab, abc, b, bc, c

C Program

```
#include <stdio.h>
#include <string.h>

void changeDateFormat(void)
{
    int day, month, year;
    char months[12][4] = {"Jan", "Feb", "Mar", "Apr", "May", "Jun",
                          "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"};

    printf("Enter date in dd/mm/yyyy format: ");
    scanf("%d/%d/%d", &day, &month, &year);

    if (month >= 1 && month <= 12)
        printf("Converted date: %02d-%s-%02d\n", day, months[month - 1], year % 100);
    else
        printf("Invalid month!\n");
}

void printSubstrings(void)
{
    char str[100];
    int i, j, k, n;

    printf("Enter a string: ");
    scanf(" %99[^\n]", str);

    n = strlen(str);
    printf("All sub-strings are:\n");

    for (i = 0; i < n; i++)
    {
        for (j = i; j < n; j++)
        {
            for (k = i; k <= j; k++)
                printf("%c", str[k]);
            printf("\n");
        }
    }
}

int main(void)
{
    int choice;

    do
    {
        printf("\n--- HDOC Day-45 Menu ---\n");
        printf("1. Change date format\n");
```

```
        printf("2. Print all sub-strings\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                changeDateFormat();
                break;
            case 2:
                printSubstrings();
                break;
            case 3:
                printf("Program ended.\n");
                break;
            default:
                printf("Invalid choice! Please try again.\n");
        }
    } while (choice != 3);

    return 0;
}
```

Sample Output

```
--- HDOC Day-45 Menu ---
1. Change date format
2. Print all sub-strings
3. Exit
Enter your choice: 1
Enter date in dd/mm/yyyy format: 12/10/2026
Converted date: 12-Oct-26

--- HDOC Day-45 Menu ---
1. Change date format
2. Print all sub-strings
3. Exit
Enter your choice: 2
Enter a string: abc
All sub-strings are:
a
ab
abc
b
bc
c
```

Functions Used

changeDateFormat() reads the day, month and year separately, uses a month-name array, and prints the year using its last two digits.

printSubstrings() uses three nested loops: the first selects the starting position, the second selects the ending position, and the third prints characters between them.

strlen() from **<string.h>** is used to find the length of the input string. **printf()** and **scanf()** are from **<stdio.h>**.

Result

The menu-driven program successfully converts the date format and prints all contiguous sub-strings of the entered string using separate user-defined functions.